EfficientNet-Image Classification Report

1. Problem Understanding

The dataset provided was highly imbalanced with very few images per class (~2–3 images). This made standard multi-class classification infeasible. The core challenge was achieving 70–80% accuracy under limited data conditions.

2. Initial Challenges

Direct brand-level classification resulted in 32 classes with extremely low samples per class, leading to poor model learning (~5–10% accuracy). Variant-level classification (blanco, reposado) also failed due to visual similarity.

3. Data Preprocessing Strategy

Multiple preprocessing strategies were explored:
- Brand-level classification: Failed due to insufficient samples.
- Subset selection: Limited scalability and reduced problem scope.
- Final Solution: Grouping strategy where brands were combined into 5 balanced classes.

This ensured sufficient samples per class and improved learning stability.

4. Final Dataset Structure

The dataset was reorganized into 5 groups to balance class distribution. Each group contained 10–20 images, ensuring meaningful learning.

5. Data Augmentation

To overcome data scarcity, augmentation techniques were applied:
- Rotation
- Zoom
- Horizontal flip
- Shifts

This artificially increased dataset diversity and improved generalization.

6. Model Architecture

EfficientNetB0 (pretrained on ImageNet) was used as the base model.
Custom classification head:
- Global Average Pooling
- Batch Normalization

- Dropout
- Dense Softmax layer

Transfer learning enabled effective feature extraction from limited data.

7. Training Strategy

Training was done in a controlled manner:
- Batch size: 8
- Learning rate: 0.001
- Early stopping to prevent overfitting
- ReduceLROnPlateau for adaptive learning

This ensured stable convergence.

8. Performance Metrics

Training Accuracy: ~71%
Validation Accuracy: ~78.5%

Loss analysis showed decreasing training loss and stable validation loss, indicating good generalization without overfitting.

9. Model Tuning

Several techniques were used:
- Reduced dropout (0.3)
- Strong augmentation
- Learning rate scheduling
- Class grouping

These collectively improved performance into the target range.

10. Conclusion

Due to dataset limitations, a grouping-based classification approach was implemented. This allowed the model to learn effectively and achieve the target accuracy range (70–80%). The solution demonstrates practical machine learning problem-solving under constraints.